# Referee report - Great Balls of FIRE IV

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We are grateful to the referee for the thorough review of our manuscript. Below we respond to each of the referee's comments, which includes details of amendments to the manuscript. All amendments are highlighted in red in the 'optional file' version of the manuscript.

* The implementation appears to overlook the influence of the galactic environment on cluster evolution, which has been taken into account in previous papers from this series. While the CMC code captures a range of tidal fields, it's unclear how this is integrated into the cluster formation process in FIREbox. Do the authors assign a fixed position within the galaxy to each cluster? Is the efficiency parameter modulated by the local conditions of the host galaxy? This needs to be explained more clearly in the text.

This remark raises a major issue in our treatment of the evolution of stellar clusters and the production of merging BBHs. We have decided not to take this aspect into account in this paper, and to save the development of a more robust semi-analytical tool for a future study. Several elements pointing out the limitations of the method presented have been added to paragraph 2.3

* In Page 5, paragraph 300: "Since the overall excess of star formation in FIREbox occurs mainly in very massive galaxies at low redshifts, it involves the formation of metal-rich stars and star clusters, which can be expected to make only a small contribution to the total population of merging BBH".

* However, the last sentence in the abstract says, "We find that these events preferentially take place in the most massive galaxies". These two statements are contradictory. I assume that the BBH that merge in massive galaxies at z=0 were formed at early stages, possibly due to an excess of SF in the early galaxies (which at the same time, have low Z). If this is the case, then the text would need to discuss the impact of the cSFR on the results. Furthermore, and in line with this, it would be interesting to show the delayed time distribution of the BBHs in these massive galaxies to those in intermediate-mass galaxies. Also the discussion on the lack of AGN feedback (and excess of star formation at high redshifts) needs to be revised in the text.

This remark reveals a lack of clarity in our analysis of the results, as well as an error in some of the numerical values presented. The last sentence of the abstract has been modified to indicate more clearly that the peak of the BBH merger rate occurs in massive galaxies **at cosmic noon**. In addition, the sentence "Since the overall excess..." is now directly followed by a mention of the more detailed discussion of the excess of star formation in FIREbox presented in 4.1. The description, in this specific subsection, of the lack of AGN feedback has been revised and the numerical values indicating the fraction of local BBH mergers originating from redshifts z¿1 have been corrected. We hope that in this version, the impact of the cosmic SFR on our predictions has been made clearer.

Finally, following your recommendation, we looked at the distribution of time delays for massive and intermediate mass galaxies, but these distributions turn out to be very similar and don't provide any particular information.

* The merger rate estimates are intrinsic and do not include any modeling of detection biases from LVK or future detectors. However, the authors directly compare their results to LVK observations, including the merger rate and primary mass distribution. If selection effects are not considered, a more detailed discussion is needed to clarify the limitations of this comparison and what aspects of the observed distributions can still be robustly interpreted.

The merger rate and primary mass distributions from the LVK Collaboration that we use here correspond to best-fit models for the **intrinsic** population of sources, thus taking into account detection biases. We have added in 3.3 a description of the nature of these curves and the limits of the comparison between these models and simulated astrophysical populations.

* The number of CMC star cluster models is relatively small, and the BBH mergers are reused across different cluster formation models. I understand the computational constraints, but this limits the stochasticity of the results. For example, the peak around 120 solar masses in the primary mass distribution appears in both the Gamma and SFRpeak models, likely due to the same underlying cluster. If this is the case, it would be interesting to describe the initial conditions of that particular cluster model. Are the BBHs formed through hierarchical mergers? How many repeated mergers are involved?

This remark on the limits of the sampling process that we used to predict the properties of a cosmological population of BBHs from massive stellar clusters is something that we were fully aware of and that we tried to quantify. This lack of stochasticity related to sampling was already discussed in 3.3, and we have added a paragraph in this subsection to reinforce this discussion. In particular, we have added a description of the physical properties of the stellar clusters that are at the origin of this peak at around 120 solar masses.

Minor comments:

* Many plots are missing error bars. Figure 1 should include error bars, and for Figure 2, I would suggest showing them in both panels or at least quantifying the uncertainties and describing them in the caption.
- Error bars have been added to Figure 1 and the legend to Figure 2 has been expanded to be more agnostic about these uncertainties.

* Figure 3: it would be much more beneficial to have a third panel with Mcl vs Zcl to complement the current figures
- third panel with Mcl vs Zcl added to Figure 3, with a description of certain features in the text (2.3)

* Section 2.2 (and subsubsections): I would define the formation rate of star clusters at the be-

ginning of section 2.2, and explain the variations of the cluster formation efficiency Gamma in the sub-subsections.
- although this comment is relevant, we have chosen to keep the structure as it is to avoid confusion between the cluster formation rate, used in the Gamma model, and the **massive** cluster formation rate, used for the other two models. A notation $\Gamma_{\rm massive}$ has been added to highlight this difference

* Some definitions are missing in the text. For instance, Mstar,90
- definition of $M_{\star,90}$ revised in 2.1

* Figure 6: the authors could make a further comparison with their results in Paper III (Fig 7).
- paragraph added in 3.3 to compare the results presented in Figure 6 here with those of Paper III